Codeflow

Experiment 01: Cleaning a Messy Dataset

Concept: Data Preprocessing

1. Start the program
2. Import pandas and Numpy libraries
3. Load the dataset
4. Display the dataset
5. check col for missing values
6. Fill missing values using mean or median
7. Convert categorical data into numerical data.
8. Display the cleaned dataset.
9. Stop the program

Experiment 02:

Concept:

1. Start the program.
2. Import required libraries.
3. Load the Iris dataset
4. Seperate input (X) and Output (Y)
5. Split dataset into 80% training and 20% testing.
6. Train the KNN model.
7. Predict the test data.
8. Calculate accuracy.
9. Predict a new flower
10. Stop the program.

Experiment 03: Predict a Number

Concept : Linear Regression.

1. Start the Program
2. Import required libraries
3. Load the car dataset.
4. Select input features
5. Select target variable (MPG).
6. Split the dataset into training and testing data.
7. Train the Linear Regression model.
8. Predict MPG values
9. Calculate R^2 score
10. Predic MPG for a new car.
11. Stop the program.

Experiment 04: Predict Yes or No

Concept: KNN and SVM classification.

1. Start the program
2. Import required libraries.
3. Load the social Network Ads dataset
4. Select input features (Age and salary).
5. Select output (Purchased).
6. Split the dataset into training and testing data.
7. Apply feature scaling.
8. Train the KNN model.
9. Train the SVM model.
10. Calculate accuracy of both models.
11. Predict whether a new customer will purchase
12. Stop the program

Experiment 05: Find Groups.

Concept: K-Means Clustering

1. Start the program.
2. Import required libraries.
3. Load the Mall Customers dataset.
4. Select Annual Income and Spending Score.
5. Apply feature scaling if used in the program.
6. Select the number of clusters ($k=5$).
7. Apply the K-Means algorithm.
8. Assign customers to clusters.
9. Display cluster groups.
10. Plot the clusters using a graphs.
11. Predict the cluster of a new customer.
12. Stop the program.

Experiment 06: Groups as a Tree

Concept: Hierarchical Clustering

1. Start the program
2. Import required libraries.
3. Load the Mall Customers dataset.
4. Select input features.
5. Apply Agglomerative Clustering.
6. Calculate distances between data points.
7. Join the closet groups.
8. Create a dendrogram.
9. Divide customers into 5 clusters.
10. Display cluster groups.
11. Predict the group of a new customer.
12. Stop the program.

Experiment 09: A Forest of Trees

Concept: Random Forest.

1. Start the program
2. Import required libraries.
3. Load the Breast Cancer dataset.
4. Seperate input and output variables.
5. Split the dataset into training and testing data.
6. Create a Decision Tree model.
7. Create a Random Forest model
8. Train both models
9. Calculate accuracy
10. Compare both models.
11. Predict whether a new tumour is benign or malignant.
12. Stop the program.

Experiment 08: Boosting (Learning from Mistakes)

Concept: AdaBoost / Boosting.

1. Start the program.
2. Import required libraries
3. Load the dataset.
4. Seperate input and output variables.
5. Split the dataset into training and testing data
6. Create the first model.
7. Identify incorrectly predict samples
8. Give more importance to incorrectly predicted samples.
9. Train the next model
10. Combine the models.
11. Calculate accuracy.
12. Predict new data.
13. Stop the program.

Experiment 09: Learn by Reward.

Concept: Reinforcement Learning (Q-Learning)

1. Start the program.
2. Import Numpy and ~~Gym~~ Gymnasium.
3. Create the FrozenLake environment.
4. Initialize the Q-table.
5. Start the training episodes
6. Agent selects an ~~op~~ action.
7. Perform the action.
8. Receive reward from the environment.
9. Update the Q-table
10. Repeat until training is completed.
11. Find the best action.
12. Display the learned path.
13. Stop the program.

Experiment 10: Neural Network

Concept: Artificial Neural Network.

1. Start the program.
2. Import Numpy, Tensorflow and required libraries.
3. Load the handwritten digits dataset.
4. Seperate input and output data.
5. Normalize the input data.
6. Split the dataset into training and testing data.
7. Build the neural network.
8. Add input, hidden and output layers.
9. Compile the model
10. Train the model using epochs
11. Evaluate the model
12. Predict a new handwritten digit.
13. Display the predicted digit.
14. Stop the program.